

# Towfiqul Islam

Senior Java Engineer | Backend Systems & DevOps | Dhaka, Bangladesh

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Portfolio: <https://towfiq-ul.github.io>

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## Professional Summary

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Senior Software Engineer with 7+ years designing, building, and maintaining Java / Spring Boot backend services and RESTful APIs for high-throughput fintech and enterprise platforms, paired with hands-on DevOps practice: Docker, Kubernetes, Jenkins/GitHub Actions CI/CD pipelines, and Azure DevOps. Most recently built core payment platform components as Augmented Staff at bKash Limited, Bangladesh's #1 Unicorn MFS, handling 2.5M+ daily transactions, while also cutting CI/CD build times ~42%. Brings additional self-directed AI/LLM engineering experience — RAG pipelines and Claude Code agentic automation tooling — directly applicable to AI-driven SDLC initiatives. Passionate about application performance, scalability, security, and reliability.

## Java / Spring Boot & DevOps Highlights

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- Built and maintained core backend services (Customer App MW, FinRec, PIN Reset MW) for bKash Limited's core payment platform, processing 2.5M+ transactions daily on Java 11–21 / Spring Boot microservices with RESTful APIs, Virtual Threads, reactive programming (WebFlux), and reactive Kafka
- Established coding standards, development workflows, and CI/CD pipelines using Jenkins, GitHub Actions, and Docker, cutting pipeline build time from 8m 29s to 4m 56s (~42% faster); mentored junior engineers on best practices and release readiness within an Agile/Scrum process
- Led the Spring Boot 2→3 migration on the PIN Reset service (javax→jakarta, AWS SDK v1→v2, Jedis→Lettuce) for zero-downtime deployment
- Delivered bKash-FinRec with corporate LDAP authentication, custom AOP-based access-control annotations, and audit-event logging shipped to a SIEM endpoint — hands-on with identity/access control and monitoring/alerting practices
- Built a Bash/shell pipeline streaming live logs from an AWS EC2 instance to developers' local machines, cutting log-check time by ~15 minutes per check
- Self-directed AI/LLM engineering: built a RAG pipeline (chunking → embeddings → vector search → grounded answers) and Claude Code agentic automation tooling — practical experience applying AI-driven automation to development and operational workflows

## Core Expertise

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- **Backend:** Java (11–21), Spring Boot, Spring MVC, Spring Security, Spring Integration, Spring Cloud, Microservices, REST APIs, gRPC, Domain-Driven Design, Event-Driven Design
- **Cloud & DevOps:** AWS (EC2, S3, SNS, SQS), Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps, CI/CD, Ansible, Linux/Unix & shell scripting, Git
- **Databases:** MySQL, Oracle, PostgreSQL, DynamoDB, Cassandra, Redis, JPA / Hibernate, Spring Data
- **Messaging:** Kafka, RabbitMQ, ActiveMQ, AWS SNS/SQS
- **Security, Identity & Observability:** OAuth2, JWT, KeyCloak, LDAP, AOP-based access control, SIEM audit logging
- **AI-Driven Engineering:** RAG pipelines, LLM integration, agentic automation tooling (Claude Code)
- **Testing:** JUnit, TestNG, Mockito, Rest Assured, Postman

## Work Experience

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### Senior Software Engineer – Exabyting

April 2024 – August 2026 | Dhaka, Bangladesh

Augmented resource at bKash Limited. Built and maintained core backend services (Customer App MW, FinRec, PIN Reset MW) on Java / Spring Boot, DynamoDB, and Kafka, processing 2.5M+ transactions daily. Established CI/CD pipelines and coding standards; mentored junior engineers on best practices and release readiness.

## Software Engineer – SELISE

November 2021 – April 2024 | Zurich, Switzerland (Remote)

Extended ThingsBoard into a multi-tenant IoT middleware platform for a Swiss enterprise customer, modeling and optimizing time-series telemetry across PostgreSQL, TimescaleDB, and Cassandra at a scale of millions of data points. Orchestrated the Swisscom integration and led direct client-facing discussions.

## Software Engineer – Exabyting

December 2019 – November 2021 | Dhaka, Bangladesh

Developed government and private Java web applications and services. Worked as partner team with bKash Limited's Solution Engineering Team. Planned development cycles, prepared release notes, and conducted deployments in an Agile/Scrum process.

## Major Java Projects

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- **bKash App** – Consumer mobile financial app; Java 11–21 / Spring Boot (WebFlux) microservices, DynamoDB, Kafka, AWS, gRPC, Kubernetes
- **bKash-FinRec** – Financial reconciliation system (BanglaQR recon) with LDAP auth, custom AOP access control, SIEM audit logging
- **PIN Reset MW** – Spring Boot SOAP+REST middleware bridging Huawei CPS, JAXB codegen from WSDL/XSD
- **Tesenso MW** – Multi-tenant IoT platform for an EV/aerospace client
- **Urstamm** – Blockchain-based supply chain system
- **FDPS** – Financial document processing with Apache Camel & Kafka
- **AMS** – Airtime Management System; automated CSV/XLS/PDF generation, SFTP operations, maker-checker approval flow, TPS-limiting service with interval-based retry

## Key Impacts

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- **bKash App**: built and maintained services processing 2.5M+ transactions/day
- **AMS**: reduced HTTP 429 rate-limit exceptions by solo-designing and building a self-contained, single-responsibility TPS-limiting service with interval-based retry, later made dynamically configurable and scalable to support Airtime Mobile Recharge Automation
- **Spring Boot 2→3 migration**: led the migration on the PIN Reset service (javax→jakarta, AWS SDK v1→v2, Jedis→Lettuce) for zero-downtime deployment
- **CI/CD**: cut build time from 8m 29s to 4m 56s (~42% faster) by implementing multithreaded build execution in the pipeline
- **Developer tooling**: built a Bash/shell pipeline streaming live logs from an AWS EC2 instance to developers' local machines, cutting log-check time by ~15 minutes per check

## Education

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B.Sc. in Computer Science & Engineering – International Islamic University Chittagong (2014 - 2019)

## Certifications & Awards

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- Java (Intermediate), C++ (Advanced), Python (Basic) – HackerRank
- IIUC IUPC 2017 – 2nd Position (Coach)
- EDU Programming Contest 2018 – 5th Position (Coach)
- CSE Week Programming Contest 2016 – 5th Position
- Solved 500+ competitive programming problems

## Open Source Contributions

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- ThingsBoard (Java, open-source IoT platform) – Pull Request #6614 (Merged)
- jmbus (Java) – Author & Maintainer; JNA wrapper around the M-Bus (Meter-Bus) protocol library libmbus, grown out of Tesenso IoT platform work
- claude-auto-resume-command – Author & Maintainer; Claude Code /auto-resume slash command plus a Stop-hook-based session guardian that auto-resumes paused sessions